# Chapter 1

# Introduction: Charting Reality with Stochastic Modeling

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Humans have been building mental models of the universe we live in since time immemorial. Within the realm of engineering and sciences, we often use modeling to mean the process of constructing a mathematical, and possibly mechanistic, description that approximates crucial aspects of reality. We call this mathematical description the "model."

The model itself is seldom the practitioner's end goal but rather the substrate for downstream tasks, such as prediction, simulation, counterfactual analysis, inference, and decision optimization. For this reason, modeling that we are interested in can further be divided into two sub-tasks:

1. *Learning* the model: to identify such a model that is well suited to produce outputs that match reality. These types of tasks often go by names such as learning, training, tuning, estimation, and inference.

2. *Using* the model: to treat the model as a proxy of reality, and use it to derive insights and recommendations to guide real-world decisions. These types of tasks often go by names such as optimization, performance evaluation, control, and policy optimization.

This class will pay special attention to what is known as *stochastic modeling.* "Stochastic" here refers to the fact that the environment that the model aims to describe is highly uncertain or volatile, and not that the model itself is random or arbitrary. Interestingly, many of the models are not themselves "random," but it is as if the more uncertain and chaotic the environment is (that is, relative to the volume of data and observations that we may have available), the simpler and more mechanistic the model becomes. The fact that probability theory, which is almost exclusively dedicated to understanding randomness itself, is almost entirely about making deterministic statements about random phenomena, can perhaps be seen as this very observation taken to its limit.

Last but not least, the word *chart* in the title is intended to carry two meanings. First, it refers to mapping reality: the process of creating understanding and orientation, as in making a map or chart. Second, and more actively, it means using that understanding to engage with, navigate, and shape reality; as in "to chart a new world" or "to chart a course through unknown waters." Just as the great navigators of old confronted the terrors of uncharted seas and transformed them into navigable frontiers, we aim to use what we know to explore and master the vast unknowns of this era of data and AI.

**Disclaimers** Let's get a few disclaimers out of the way before we begin: The frameworks, approaches, and viewpoints in these notes will contain plenty of subjective opinions and personal biases, influenced by my own research and industry experiences. It is my hope that some of this subjectivity helps animate the discussion, but obviously, please take these views as such, with appropriate skepticism.

Because of the above, I will do my best to make the parts of the notes about historical events and empirical or theoretical research findings stand apart from the more subjective remarks.

Some chapters may be grounded in specific papers, in which case I may talk about the results and findings directly without necessarily reproducing the content here. I will assume that the reader has the relevant paper(s) at their disposal; if something seems missing from these notes, it is probably discussed in depth in the source text.

## 1.1 From Reality to Model

What is "reality", and what does it mean for a model to "capture it"? I won't even attempt to do justice to these age-old questions, but rather will simply try to provide some practical definitions that will suffice for subsequent concepts to build on.

For our purpose, you can think of reality as simply a pair of variables in some abstract spaces, $(x, y)$. The law governing the relationship between $x$ and $y$ is the central objective of interest. A model, therefore, is a functional mapping from some input $x$ to some (presumed) output $y'$. A good model is one where $y'$ resembles the actual $y$ in reality by some metric.

In predictive modeling, $x$ typically represents the input information that we know, and $y'$ the output information that we don't yet know. However, this distinction between the known vs. unknown information is not necessarily the focal point. In many scientific inquiries, for instance, one may simultaneously observe the $x$ (e.g., locations of the planets) and the $y$ (e.g., their orbits and velocities), and the quest is rather to come up with mechanistically elucidating explanations for their relationships that are as empirically accurate as they are mechanistically convincing. In fact, in science, what I call in these notes "model" is more often referred to as "theory."

Below is a slightly more formal definition, which also allows for the mapping to incorporate some randomness, should the practitioner choose to do so.

**Definition 1.1.1** (Model & Model Family)**.** Let $\mathcal{X}$ be the set of all possible inputs, and let $\mathcal{Y}$ be the set of all possible outcomes. Let $Z$ be the random seed, a uniform random variable taking values in $[0, 1]$.

A model is a function $f$ that maps $x \in \mathcal{X}$ and $z \in [0, 1]$ to an element in the outcome set $\mathcal{Y}$, where we use $f(x, z)$ to generate the outcome given input $x$. A deterministic model is one in which the output does not depend on the random seed, i.e., $f(x, z) = f(x, z')$ for all $z, z' \in [0, 1]$. A set of functions that could plausibly contain a desirable model, $\mathcal{F}$, is called a model family. ◇

By this point, the reader could probably sense that despite the appearance of logical and mathematical rigor, there is still going to be quite some room for subjectivity in one's modeling journey. How to choose the model family? When should one use randomness? Who is to define what "accurate" means, and what impact it would have on the type of model selected, etc... As we will see, these suspicions are well warranted. A few more observations are in order:

Inputs can further be divided into those that are *uncontrollable*, like weather or temperature and those that are *controllable*, such as driving instructions or prices. We sometimes call the uncontrollable input by names like *parameters* and *problem primitives*, and the controllable input *decision variables* or *interventions*. Both forms influence the output through the model $f$, but as we will see, they have very different and significant downstream ramifications on operations, policy design and data collection.

It's worthless to define a very general family of models but without the ability to make use of a single one. As such, not only should the model family be rich and comprehensive enough to capture reality, we also want the models to be *practical*, which means that they can be reliably found, efficiently computed, and ideally, intuitively compelling.

*Exercise* 1.1. Can you come up with an example of models and model families in the following domains?

1. Physics: planetary motion, mechanics, thermodynamics...
2. Service systems: queues, congestion, marketplaces...
3. Machine learning and artificial intelligence: prediction, learning theory, transformers...

## 1.2 Frame, Learn, Use, Evaluate – The Modeling Workflow

*Frame, Learn, Use, Evaluate*

We can now revisit what a typical end-to-end modeling workflow looks like, which involves the following steps:

1. **Framing**: To determine the model family.
2. **Learning**: To identify the model within the model family.
3. **Using**: To use the model for optimizing downstream decisions, or for conceptual understanding.
4. **Evaluating**: To assess whether the preceding steps yield results that align with or satisfactorily reflect empirical reality.

**Framing and Learning** We have already talked about the Learning and Using phases at the beginning of this chapter. One may wonder why not just lump the Framing step into Learning; after all, isn't choosing the model family simply part of learning and fitting the model?

Framing is a fascinating step of the process and at times glossed over by practitioners. Being the step that pins down the universe of models that one will ever consider, it is not difficult to imagine that choosing the right model family (e.g., linear models, neural nets, queueing models, etc.) has a profound downstream impact. But for all its importance, there does not appear to be any purely objective procedure to guide Framing. It is rather a labor of subjective judgment, expert hunches, and, at times, strokes of sheer divine insight (think Einstein and Newton). Last but not least, this Framing step is strictly unskippable, as articulated by the No Free Lunch theorem of learning theory, which we examine next.

**Using** It is worth noting that the Using step encapsulates both using the model to influence specific actions, such as in engineering and operations research, and using it to generate conceptual understandings without immediate downstream actions, such as in science. For taking specific actions, the practitioner typically guides the action by maximizing a certain objective function or reward function that encodes their preferences. Interestingly, one could argue that this is true even when the practitioner's goal is merely to derive conceptual understanding: in this case, they may want to generate explanations of phenomena that score favorably against a certain implicit value function that encodes what is perceived to be elegant or beautiful or convincing by the standard of the respective scientific audience.

**Evaluating** The goal of evaluation is simply to understand whether modeling was performed successfully and to measure how successful it is. But it's important to understand that evaluation should be performed in both a local and a global sense. By local, I mean evaluating the accuracy and soundness of the procedure in each of the steps: whether a model fit was performed to a sufficient level of accuracy, or if an optimization algorithm was run correctly to identify an optimal solution. These types of local validation should be performed in each of the modeling steps.

By global, I mean whether the entire modeling approach or procedure was suited for, or matches with, empirical reality. A very common pitfall is that the practitioner, upon observing the success of local evaluations, may not realize that the global evaluation is still necessary. For example, someone might build a model and successfully estimate its parameters with very low error. Locally, the fitting procedure seems flawless, but correctly estimating parameters in a model does not mean the modeling approach itself is correct. If the real-world dynamics look nothing like the chosen model, the entire approach may still fail once deployed. Therefore, perhaps counterintuitively, a good global evaluation should be approached from an "adversarial", or at least "orthogonal", angle: it cannot be reliant upon the very modeling assumptions that we are supposed to evaluate to begin with; it must be tested against independent, empirical reality.

What this all could mean in practice, for instance, is that however theoretically compelling or impressive the modeling framework one may have used in *designing* a new algorithm or product, the said product must be put through a model-agnostic, end-to-end *experiment*, such as a Randomized Controlled Trial (RCT) or Switchback experiments, to assess the real-world effectiveness.

**Stages Interact** Most of the topics in this course will involve at least one of these steps, and sometimes multiple steps simultaneously. The four steps also have intricate interactions and feedback loops. For instance, it is perhaps obvious that knowing the objective function (which, ironically, is anything but "objective") of the decision problem in Step 3 can heavily influence Framing and Learning. In another example, the Learning and Using steps may not always take place in a clean-cut, sequential manner, but can be temporally interleaved, where the practitioner makes decisions that serve both to learn and to act; we will see many such examples in active learning and reinforcement learning.

*Exercise* 1.2. Can you recall a recent instance in your own work or in the work of others where Framing was used? What was the model family, and how did you/they select it?

## 1.3 Why Framing Is Unskippable: The No Free Lunch Theorem

But before we continue, let us try to answer a more basic question: does good modeling really require the ingenuity of smart people like Newton, Erlang, or Einstein? Put differently, can we bypass Framing by fixing an extremely rich model family and relying on Learning alone to select a predictor from data? If so, the role of human intervention would be much smaller.

The No Free Lunch theorem provides an informative answer: that it depends on how much data you have. This strategy breaks down when sample size is small relative to the complexity of the stochastic environment, and when this happens, some form of inductive bias (largely done by humans at least until the onset of AI) via Framing remains necessary.

Let $\mathcal{X}$ be the input domain and $\mathcal{Y} = \{0, 1\}$ the binary outcome set. Let $m$ be a positive integer denoting sample size. In the notation of this chapter, we focus on deterministic models, so a model is a function $h : \mathcal{X} \to \mathcal{Y}$. A learning algorithm $A$ maps a training sample $S = ((x_1, y_1), \dots, (x_m, y_m)) \in (\mathcal{X} \times \mathcal{Y})^m$ to a model $A(S) : \mathcal{X} \to \mathcal{Y}$. The sample is drawn i.i.d. from a distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, denoted $S \sim \mathcal{D}^m$. For any model $f$, its population 0–1 loss is

$$L_{\mathcal{D}}(f) \;=\; \mathbb{P}_{(X,Y)\sim\mathcal{D}}(f(X) \neq Y)\,.$$

The following version of the theorem is from (Shalev-Shwartz and Ben-David, 2014, §5.1).

*Why Frame? The No Free Lunch Theorem*

**Theorem 1.3.1.** *Let $A$ be any learning algorithm for binary classification over the input domain $\mathcal{X}$. Let $m < |\mathcal{X}|/2$ be the training set size. Then there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0, 1\}$ such that:*

1. *There exists a function $f : \mathcal{X} \to \{0, 1\}$ with $L_{\mathcal{D}}(f) = 0$.*
2. *With probability at least $1/7$ over the choice of $S \sim \mathcal{D}^m$, we have*

$$L_{\mathcal{D}}(A(S)) \geq 1/8.$$

This theorem says that for every learner, there exists a learning task on which it fails, even though that same task can be learned successfully by another learner (possibly with additional prior knowledge). The proof intuition is simple: pick a subset $C \subseteq \mathcal{X}$ of size $2m$. A learner trained on $m$ examples sees labels for only about half of $C$, leaving the unseen half unconstrained. One can then construct a target labeling on those unseen points that contradicts the learner's output, producing nontrivial generalization error.

Because we are in a fixed binary output space, the number of possible mappings from $\mathcal{X}$ to $\mathcal{Y}$ grows with $|\mathcal{X}|$, which serves as a proxy for environment complexity. The theorem then implies that when sample size is small relative to this complexity, we cannot expect Learning alone to succeed reliably.

More generally, when data is small relative to environment complexity, "choose the richest family and fit" is not a viable strategy: if the family is too large, fitting is underdetermined and overfitting becomes unavoidable; if it is too small, we lose representational power. Reliable learning therefore requires inductive bias, supplied through Framing choices often informed by domain knowledge and human judgment.

**But is reality so unkind?** Theorem 1.3.1 can sound pessimistic because it is worst-case: the environment is chosen adversarially against the learner. Just because a learner *can* fail does not mean it is likely to fail on naturally occurring tasks. The practical success of very large AI models is a reminder not to dismiss large model families too quickly. Even then, however, framing choices remain deeply present in architecture, objective design, data curation, and evaluation protocols.

*Exercise* 1.3. What would be a rough estimate of the size of the model family for a current-gen large language model? You may use some back-of-the-envelope heuristics, such as quantizing parameters. Do you think the No Free Lunch theorem applies in that setting? What do you notice?

## 1.4 The Blessing and Curse of Mechanistic Interpretability

What makes a model good beyond accuracy? Suppose multiple models or model families could yield similar accuracies in capturing the laws of the input-output relationships in observed data. Should we in that case prefer some models over the others by virtue of other characteristics? ... Debates such as these often arise in practice, and usually someone brings up the consideration of interpretability.

In modeling, *interpretability* refers to the property that the selection or behavior of a model be relatively easy to explain to a human while resorting to basic principles. This is obviously not a great definition, because what it means to explain is also highly subjective. An average student of the quantitative sciences today would likely agree that Newtonian laws such as $F = ma$ and their various predictions are more "intuitive" and "interpretable," than, say, the billions of weights arrived at from pre-training a deep neural network.

*Mechanistic modeling*

For stochastic modeling, my observation has been that researchers and practitioners seem to associate interpretability with models that are *mechanistic* in the following sense:

1. The overall model can be complex in the sense that it consists of many interacting building blocks.
   *That is, complexity can arise from the interaction of simple sub parts.*

2. The interactions between these building blocks should be relatively transparent and simple.
   *That is, you cannot cheat by hiding complexity in their interactions.*

3. Each building block can be a functional unit, or a certain mathematical claim. However, their mechanism or function should be very mathematically transparent or easily empirically verifiable.
   *That is, you also cannot cheat by hiding complexity in the inner working of the basic elements.*

Modeling approaches that conform to the above style are mechanistic in the same sense a watch is: the overall mechanism and emerging behavior of a mechanical watch can be extremely complex and mysterious from the outset, but if you take the watch apart, the nature of the individual components and how they interact are (relatively) easily understood and explained by basic material science or physics. And just like a watchsmith, some researchers may feel that one does not truly "understand" something unless they are able to take a model apart into individual components, describe the functions of the basic components and how they interact, before putting them back together to explain the emergent properties of the whole model itself.

Each era of scientific discourse dating back to antiquity seemed to have its own preferred norm of mechanistic interpretability for that respective era, including our own. This is perhaps quite understandable in the sense that humans instinctively rely on such mechanistic understanding to assure themselves that their model of the world would generalize well in a turbulent environment despite being learned from limited observations or data (That is, we are not overfitting the limited observations using some "wacky theory").

But what is perhaps more interesting is that paradigm-shifting scientific advancements were often made by people who experienced precisely the opposite: they come up with new models and theories that offer more accurate, or simply novel, explanations of the world, even if such theories and models were not mechanistically interpretable at the time of their invention.

### 1.4.1 The Example of Copernican Revolution

This dynamic ebb and flow around mechanistic beauty is nowhere more evident than in one of the most famous examples of modeling exercises in scientific history: the theory of the movements of the planets and the dawn of Newtonian physics.

Prior to Copernicus's heliocentric model in 1543, the geocentric Ptolemaic system was the dominant theory that explains and predicts the movements of celestial bodies. The Ptolemaic system posits that all planets and stars, including the Sun, revolve around Earth. It has also developed intricate theories that accurately predict their locations and trajectories.

When Copernicus's heliocentric model first came out, it was conceptually transformative but initially less accurate in prediction than the more mature Ptolemaic system. With hindsight, we could say that Copernicus' new theory was a far more "correct" mechanistic model of reality, but at the expense, at least initially, of fidelity and accuracy.

Amazingly, the subsequent development would prove to flow in the opposite direction, where predictive accuracy came at the expense of mechanistic interpretability. Aided by Tycho Brahe's precise naked-eye (!) observations and working as Tycho's former apprentice, Kepler created his famous laws of elliptical planetary motion: that planets revolve around the Sun along elliptical trajectories, with the Sun at one focus, etc. Kepler's theory was remarkably accurate at fitting the observational data and was key in making Copernicus's heliocentrism decisively empirically superior to that of the earth-centric theory. But this giant leap is in fact not so mechanistically interpretable: it does not explain *why* planets move in an ellipsoid, but simply postulates that it *does*.

Newton's theory of gravitation decades later would appear to bring the entire story into a complete, triumphant resolution. From Newton's formula of gravitational force between two bodies, $F = \frac{Gm_1m_2}{R^2}$, we can logically deduce why planets revolve around the Sun following elliptical trajectories, and indeed all of Kepler's laws. We no longer have to just assume that planets go around the Sun in ellipsoids, but we now feel that we can better explain why it does so.

What's more remarkable is that not only does the Newtonian theory seem more interpretable, it is also better at making more accurate predictions. Indeed, planets in the solar system do not in fact move in perfect ellipsoids due to the gravitational interactions with other planets; they only appear to approximately follow an elliptical trajectory because the pull from the Sun is so dominant. All of this was not known, and could not have been reasoned through, without Newton's theory of gravity.

A modern reader might have naturally concluded that Newton had achieved the perfect union that most scientists could only dream of: a theory that's both more mechanistically sound and empirically superior. Shockingly, Newton in fact considered his theory of gravity precisely a result of being willing to go *outside* of the bounds of mechanistic interpretability in exchange for superior empirical success. In his *Principia*, he wrote:[1]

*Newton's philosophy*

> *I have not as yet been able to deduce from phenomena the reason for these properties of gravity, and I do not feign hypotheses; for whatever is not deduced from the phenomena must be called a hypothesis, and hypotheses, whether metaphysical or physical, whether of occult qualities or mechanical, have no place in experimental philosophy. In this experimental philosophy, propositions are deduced from the phenomena and made general by induction. It is enough that gravity really exists and acts according to the laws that we have set forth and is sufficient to explain all the motions of the heavenly bodies and of our sea.*

In particular, Newton did not try to explain the true nature of the "gravitational pull." Where does gravity come from? What carries it? What are its fundamental elements?... Newton simply used it practically, as a theoretical device to explain and "fit" reality. In fact, he goes further by arguing that it is precisely his willingness to first leverage this conceptual device "gravity" to build the model, and worry about its mechanistic nature later, that he was able to succeed at creating a theory that is vastly more empirically powerful.

Mapping Newton's theory back to the criteria we showed in the beginning of this section, one would argue that the concept of gravitational force as well as the gravitational force formula serve as fundamental building blocks, from which complex models of celestial motions can be built. I suspect that Newton's theory *feels* so mechanistic today not because its foundational building blocks are necessarily themselves mechanistically explainable, but rather that they have been so intensely empirically verified and the theory so widely taught, that such conceptual devices and their expected properties have been absorbed into our collective consciousness and scientific consensus as sufficient for serving as unquestioned building blocks.

*Mechanistic interpretability vs. empirical success*

Taking stock of all that had happened, the modeling progression in the Copernican revolution doesn't seem to be as simple as the linear narrative we learned in grade school, but a far more nuanced interplay between mechanistic interpretability and empirical success.

Copernicus created a theory that is more mechanistically correct but less empirically accurate. Kepler builds on Copernicus' theory by adding a layer that is *not* so mechanistically interpretable, but yielding much more empirical success. Newton provides a theory that is both more mechanistically appealing and more empirically accurate than Kepler's. However, upon closer inspection, even Newton's theory relies on *building blocks* that are not themselves mechanistically explained, and Newton himself attributed parts of his success for precisely being willing to let go of mechanistic interpretability.

[1] *Principia*, Book III, General Scholium (added in the 1713 second edition; retained in the 1726 third edition). English translation in (Newton, 1999, p. 943).

*Exercise* 1.4.

1. Can you think of an example of a mechanistically interpretable model family, versus one that is not?
2. Do you think practitioners innately prefer interpretable models? Why, or why not?
3. Do you think mechanistic interpretability is still relevant today despite the advent of massive datasets and AI models? If so, in what domains do you think they are more relevant than others? Why?
4. Do you think interpretability matters in building AI or AI-powered systems?

## 1.5 Summary

We have laid out in this chapter a first-principles map of what stochastic modeling is ultimately about: not model-building as an end in itself, but as a means for learning, prediction, explanation, and ultimately better decision making. We introduced a minimal formal language for models and model families, while acknowledging that modeling remains irreducibly shaped by judgment: what counts as a good model depends on purpose, tractability, and the value function, explicit or implicit, which governs what we care about.

We organized the workflow into four stages: **Framing**, **Learning**, **Using**, and **Evaluating**. Framing determines what model families we are even willing to consider. Learning identifies a model within that family. Using turns the model into decisions or understanding. Evaluating assesses both the local soundness of each step and the global empirical validity of the overall approach. In practice, these four stages interact and feed back on one another.

We then used the No Free Lunch theorem to make precise why Framing is unskippable: when data is limited relative to environment complexity, inductive bias is not optional. Finally, we discussed the blessing and curse of mechanistic interpretability. Through the Copernicus-Kepler-Newton story arc, we saw that progress is not always a synchronized, straight march toward both interpretability and empirical accuracy. Sometimes one comes first, then the other.

## 1.6 Recommended Reading

For the historical narrative around Copernicus, Kepler, and Newton in this chapter, see the discussion in (Gingerich, 2007; Kuhn, 1957) and the original statement of Kepler's planetary laws in (Kepler, 1609). For Newton's gravitational framework and his methodological stance in *Principia*, see (Newton, 1687, 1999). The discussions around the blessing and curse of mechanistic interpretability also draw inspiration from excellent writings in *The Knowledge Machine* and *The Bitter Lesson* (Strevens, 2020; Sutton, 2019). The No Free Lunch result stated in Theorem 1.3.1 can be found in Section 5.1 of (Shalev-Shwartz and Ben-David, 2014), along with its proof.

## Bibliography

Amazon Web Services (2024). Amazon Bedrock Service Level Agreement. AWS SLA page. Accessed 2024.

Armony, M., Israelit, S., Mandelbaum, A., Marmor, Y. N., Tseytlin, Y., and Yom-Tov, G. B. (2015). On patient flow in hospitals: A data-based queueing-science perspective. *Stochastic Systems*, 5(1):146–194.

Baseten (2026). How Hebbia Uses Baseten to Power AI Workflows. Baseten customer story. Discusses TTFT-sensitive inference workloads; accessed April 10, 2026.

Bhat, U. N. (1969). Sixty years of queueing theory. *Management Science*, 15(6):B280–B294. Application Series.

Box, J. F. (1978). *R. A. Fisher: The Life of a Scientist*. John Wiley & Sons, New York.

Brown, L., Gans, N., Mandelbaum, A., Sakov, A., Shen, H., Zeltyn, S., and Zhao, L. (2005). Statistical analysis of a telephone call center: A queueing-science perspective. *Journal of the American Statistical Association*, 100(469):36–50.

Danmarks Statistik (1901). Folkemængden 1. februar 1901 i Kongeriget Danmark efter de vigtigste administrative inddelinger. Danmarks Statistik publication page. Official census publication; accessed April 2, 2026.

Encyclopaedia Britannica (2026a). Jerzy neyman. https://www.britannica.com/biography/Jerzy-Neyman. Accessible biography; states Neyman was appointed statistician at the Institute of Agriculture in Bydgoszcz, Poland; accessed 2026-04-20.

Encyclopaedia Britannica (2026b). Loading coil. Britannica article page. Overview of loading-coil purpose in long-distance telephony; accessed April 2, 2026.

Erlang, A. K. (1909). Sandsynlighedsregning og telefonsamtaler. Digital scan (Royal Danish Library). [Digital scan; accessed 2026-03-29].

Erlang, A. K. (1917). Solution of some problems in the theory of probabilities of significance in automatic telephone exchanges. *Elektroteknikeren*, 13:5–13. Original Danish title: Łosning af nogle Problemer fra Sandsynlighedsregningen af Betydning for de automatiske Telefoncentraler. Archived scan: Wayback copy.

Fischer, C. S. (1978). The spatial diffusion of information and the long-distance telephone, 1875–1900. *Social Science History*, 2(2):173–196.

Fisher, R. A. (1935). *The Design of Experiments*. Oliver and Boyd, Edinburgh.

Gingerich, O. (2007). The copernican myths. https://physicstoday.aip.org/features/the-copernican-myths. [Physics Today; accessed 2026-03-22].

Gross, D., Shortle, J. F., Thompson, J. M., and Harris, C. M. (2008). *Fundamentals of Queueing Theory*. Wiley, Hoboken, NJ, 4 edition.

Hajek, B. (2015). *Random processes for engineers*. Cambridge university press. https://hajek.ece.illinois.edu/ECE534Notes.html.

Hebbia (2025). The Distributed System Behind Hebbia's High-Scale AI. Hebbia engineering blog. Published March 17, 2025; accessed April 10, 2026.

Holland, P. W. (1986). Statistics and causal inference. *Journal of the American Statistical Association*, 81(396):945–960.

INFORMS (2026). Who Was Agner Krarup Erlang? INFORMS Erlang page. INFORMS Erlang Prize background page; accessed April 2, 2026.

Jewett, F. B. (1951). The evolution of inductive loading. *Bell System Technical Journal*, 30(2):123–140.

Kendall, D. G. (1953). Stochastic processes occurring in the theory of queues and their analysis by the method of the imbedded markov chain. *The Annals of Mathematical Statistics*, 24(3):338–354.

Kepler, J. (1609). *Astronomia Nova*. Heidelberg. Digitized edition metadata via e-rara.

Kuhn, T. S. (1957). *The Copernican Revolution: Planetary Astronomy in the Development of Western Thought*. Harvard University Press, Cambridge, MA.

Li, S., Johari, R., Kuang, X., and Wager, S. (2023). Experimenting under stochastic congestion. arXiv preprint arXiv:2302.12093 (v5 updated 2026-02-15).

National Park Service (2005). New Jersey Bell Headquarters Building, National Register of Historic Places Registration Form. NPS/NJ historic register PDF. Includes historical summary of the Independent Telephone Movement and entry scale; accessed April 2, 2026.

New York Public Library (2026). 1964–1965 new york world's fair corporation records, 1939–1966. NYPL archives entry. Photo archive; accessed 2026-03-30.

Newton, I. (1687). *Philosophiae Naturalis Principia Mathematica*. Jussu Societatis Regiae ac Typis Josephi Streater, London.

Newton, I. (1999). *The Principia: Mathematical Principles of Natural Philosophy*. University of California Press, Berkeley.

Neyman, J. (1923). On the application of probability theory to agricultural experiments. essay on principles. section 9. *Statistical Science*, 5(4):465–472. English translation published in 1990.

Neyman, J. and Scott, E. L. (1967). Statistical aspect of the problem of carcinogenesis. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, Volume IV: Biology and Problems of Health*, pages 745–776, Berkeley and Los Angeles. University of California Press.

Our World in Data (2026a). Mean years of schooling (long-run). OWID dataset link. Long-run education dataset; accessed April 2, 2026.

Our World in Data (2026b). Share of the population with tertiary education. OWID dataset link. Dataset adapted from Barro and Lee (2015) and Lee and Lee (2016); accessed April 2, 2026.

Ross, S. M. (2014). *Introduction to Probability Models*. Academic Press, Amsterdam, 11 edition.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701.

Rubin, D. B. (1980). Randomization analysis of experimental data: The fisher randomization test comment. *Journal of the American Statistical Association*, 75(371):591.

Shalev-Shwartz, S. and Ben-David, S. (2014). *Understanding machine learning: From theory to algorithms*. Cambridge university press.

Sherbrooke, C. C. (1968). Metric: A multi-echelon technique for recoverable item control. *Operations Research*, 16(1):122–141.

Sherbrooke, C. C. (2004). *Optimal Inventory Modeling of Systems: Multi-Echelon Techniques*. Kluwer Academic Publishers, Boston, 2 edition.

Skee, J. (2019). The history of the analysis of queues and waiting lines. INFORMS history page. Edited by Linus Schrage; INFORMS History of O.R. Excellence, Queueing Models; accessed 2026-03-30.

Skee, J. D. (2016). *By the Numbers: Confidence, Consultants, and the Construction of Mass Leisure, 1953–1975*. Phd dissertation, University of California, Berkeley.

Strevens, M. (2020). *The Knowledge Machine: How Irrationality Created Modern Science*. Liveright Publishing Corporation, New York.

Sutton, R. (2019). The bitter lesson. http://www.incompleteideas.net/IncIdeas/BitterLesson.html. [Online essay; accessed 2026-03-26].

Ulrich, R. and Miller, J. (1993). Information processing models generating lognormally distributed reaction times. *Journal of Mathematical Psychology*, 37:513–525.

U.S. Bureau of Labor Statistics (2026). Cpi inflation calculator. https://www.bls.gov/data/inflation_calculator.htm. [CPI-U based; accessed 2026-04-02].

U.S. Census Bureau (2023). Facts by Decade: 1900. U.S. Census page. Includes 1900 urban population table; accessed April 2, 2026.

Wager, S. (2024). Causal inference: A statistical learning approach. Stanford Graduate School of Business course notes. [Referenced in Chapter 3].

Whitt, W. (1993). Approximations for the gi/g/m queue. *Production and Operations Management*, 2:114–161.

Xu, K. and Chan, C. W. (2016). Using future information to reduce waiting times in the emergency department via diversion. *Manufacturing & Service Operations Management*, 18(3):314–331.